Project summary: Adjoint-based Maximisation of Heat Flux in Models of Stellar Convection

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Introduction

Convection is an important natural process that governs the transport of energy through stellar interiors, influencing the structure, luminosity, and lifetime of stars. To break free of the assumptions made by Mixing Length Theory (MLT), we can use Dedalus v3 [1] to understand which flow structures are most efficient at transporting heat within Rayleigh-Bénard systems.

Simulating these highly turbulent astrophysical environments requires sophisticated mathematical and computational modelling. By optimising systems of Rayleigh-Bénard convection, we aim to maximise the amount of convective heat transfer within the system, parametrised by the Nusselt number, Nu, and find an appropriate scaling between this Nusselt number and the Rayleigh number, which gives a measure of the strength of thermal driving in the system. Various papers have explored methods to approximate this scaling; many use different boundary conditions or methods to obtain the maximal Nusselt number [2][3]. This project aims to provide another approach to approximate this scaling law using adjoint-based optimisation methods [4][5][1].

Asymptotic Scaling Law

Using the widely-accepted Boussinesq approximation, we can construct both the forward and adjoint problems solved to iterate through domain widths (full proof can be found on GitHub). The spectral solver first solves the forward (physical) problem, uses this final state as the initial state of the adjoint system, and solves the adjoint system backwards in time. This is used to calculate the gradient of the Nusselt number with respect to the domain width, which is then applied to the gradient-ascent algorithm to iterate towards the optimal domain width.

By iteratively applying domain-optimisation across a wide sweep of Rayleigh numbers, we can extract the fundamental empirical scaling. The maximum achievable Nusselt number for free-slip boundaries follows a remarkably clean fractional scaling law (see Figure 1):

$$Nu \sim 1 + \frac{1}{7} Ra^{3/8} \tag{1}$$

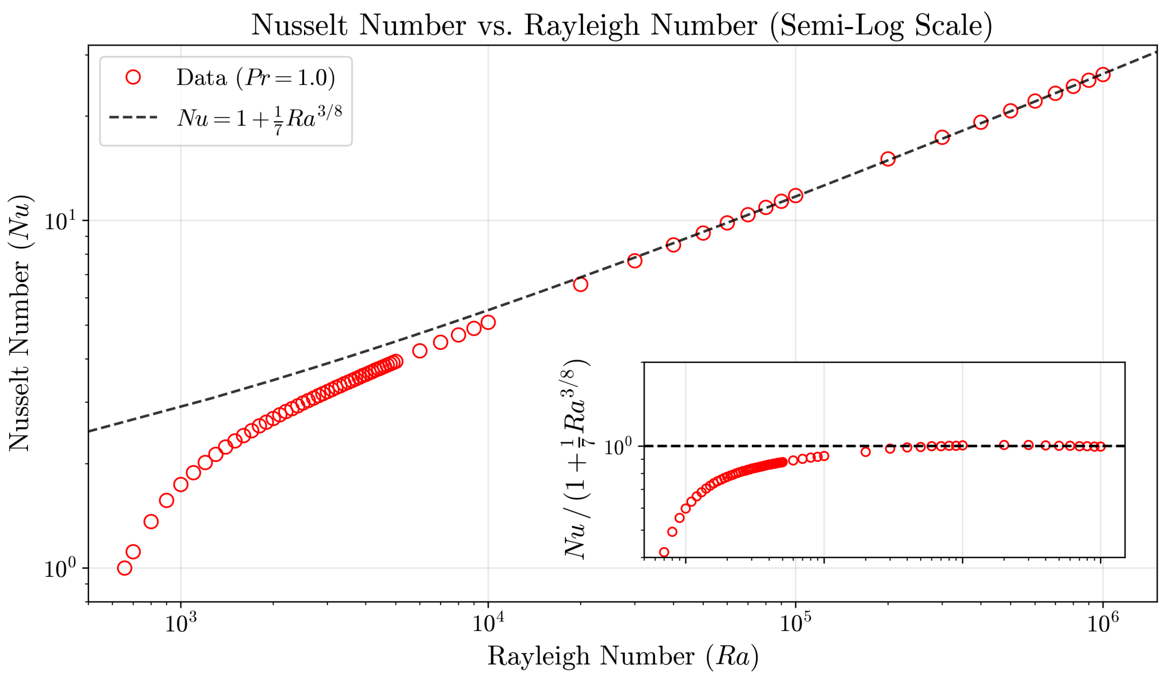

Figure 1: Optimal Nusselt number against Rayleigh number under free-slip boundary conditions.

The Nusselt number can be a measure of the potential energy of the fluid. To maintain a steady state, this energy must be expended on overcoming internal friction, measured globally as viscous dissipation, ϵ. Under free-slip conditions, our simulations confirm this exact energy balance [6]:

$$\epsilon = \sqrt{RaPr}(Nu - 1) \tag{2}$$

Consequently, geometrically optimising the domain to maximise convective heat transfer inherently forces the fluid to maximise its global viscous dissipation.

Conclusions

In this project, we successfully demonstrated the power of adjoint-based optimisation to maximise convective heat transfer within Rayleigh-Bénard systems rigorously. By shifting away from traditional brute-force parameter sweeps and leveraging the continuous adjoint equations alongside a spectral solver, we established a highly efficient computational framework for discovering optimal fluid states.

By optimising the spatial domain width, we minimised turbulent contributions from higher-order Fourier modes, maximising global viscous dissipation. Finding this optimal state allowed us to isolate a precise, theoretical asymptotic scaling law for free-slip boundaries, $Nu \sim 1 + \frac{1}{7}Ra^{3/8}$. While work still needs to be done to fully place this scaling in the context of the vast literature, this relationship offers a powerful predictive tool for extrapolating the behaviour of stellar media in extreme high-Rayleigh regimes that currently sit beyond the limits of practical direct numerical simulation.

In this project, we demonstrated our ideas using the paradigm problem of Rayleigh–Bénard convection. Natural next steps would be to incorporate the effects of rotation, magnetic fields, and fluid compressibility, thereby moving towards more realistic models of stellar interiors.

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References

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